Introducción al modelo OMOP-CDM

Sesión 5.- Publicación y explotación

Alberto Labarga – Febrero 2026

I am Alberto

alabarga

alabarga

/in/albertolabarga

alberto.labarga@gmail.com

OMOP-CDM
El curso

El curso tiene como objetivo proporcionar una comprensión profunda del modelo de datos **OMOP-CDM** de la OHDSI, incluida su **estructura**, tablas clave y **vocabularios**.

Abordaremos también como **transformar** datos al modelo y explorar y analizar la **calidad** de los datos transformados.

Febrero 2026

L	M	X	J	V	S	D
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	1
2	3					

Objetivos del curso

01 Introducción al modelo OMOP-CDM, a sus tablas y sus vocabularios

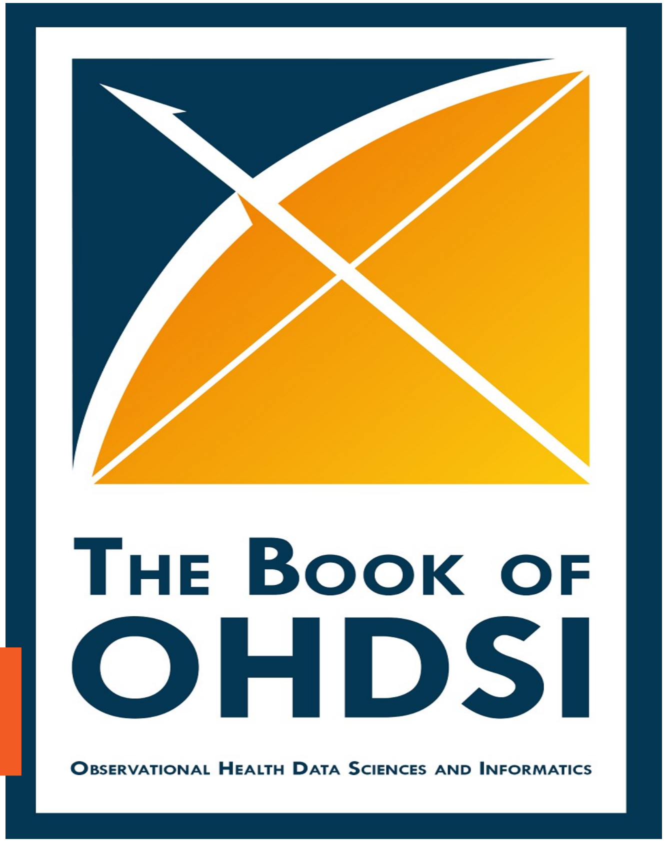
02 Transformación de datos al modelo OMOP-CDM utilizando la suite de herramientas de la OHDSI

03 Análisis de calidad y explotación de una base de datos OMOP-CDM

Recursos de aprendizaje

Sitio web: https://github.com/alabarga/curso-omop-cdm

Libro https://ohdsi.github.io/TheBookOfOhdsi/



FHIR & OMOP

de datos genómicos

OHDSI

OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

The Observational Medical Outcomes Partnership Common Data Model (**OMOP-CDM**) ha sido desarrollado por y para la comunidad científica, proporcionándo las herramientas para generar evidencia de alta calidad y reproducible a partir de datos de salud observacionales. Los datos pueden ser extraídos, transformados y cargados directamente desde los sistemas de información clínica.

FHIR

FHIR (Fast Healthcare Interoperability Resources) es un estándar universal para el intercambio de registros de salud de pacientes en sistemas de historia clínica electrónica. Está compuesto por lo que se denominan recursos, agrupados en perfiles que especifican cómo debe representarse un elemento de información particular.

The OMOP on FHIR is a Java implementation of a FHIR Server (using HAPI FHIR) built on top of an OMOP CDM Database, providing a mapping layer in-between. de información clínica.

FHIR to OMOP FHIR Implementation Guide

A FHIR implementation guide that supports conversion of data from FHIR to OMOP and OMOP to FHIRThis standard provides details on how to transform healthcare data from FHIR to the OMOP Common Data Model. It aims to bridge the gap between these two widely used formats in healthcare and research. The standard defines mappings between FHIR resources and OMOP data tables, focusing on commonly used EHR data.

OHDSI*

EL ECOSISTEMA DE OMOP-CDM


FAIR

FAIR
DATA

FINDABLE

Identificadores únicos y metadatos que pueden usarse para encontrar los datos de manera sencilla

ACCESSIBLE

Los datos son abiertos, libres y universalmente disponibles para la investigación

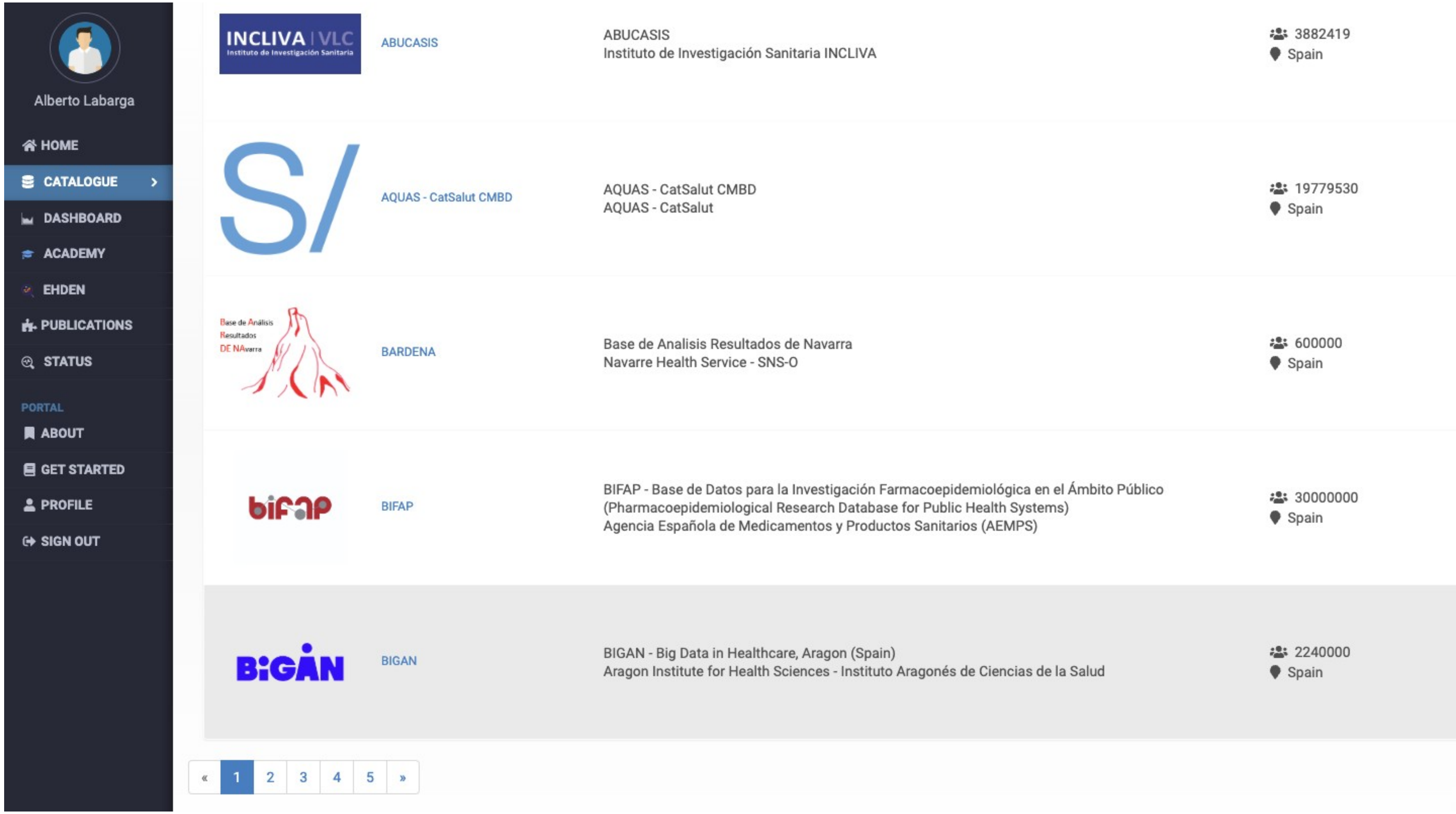
INTER-OPERABLE

Los datos utilizan formatos libres y estándar utilizados por un amplio número de aplicaciones

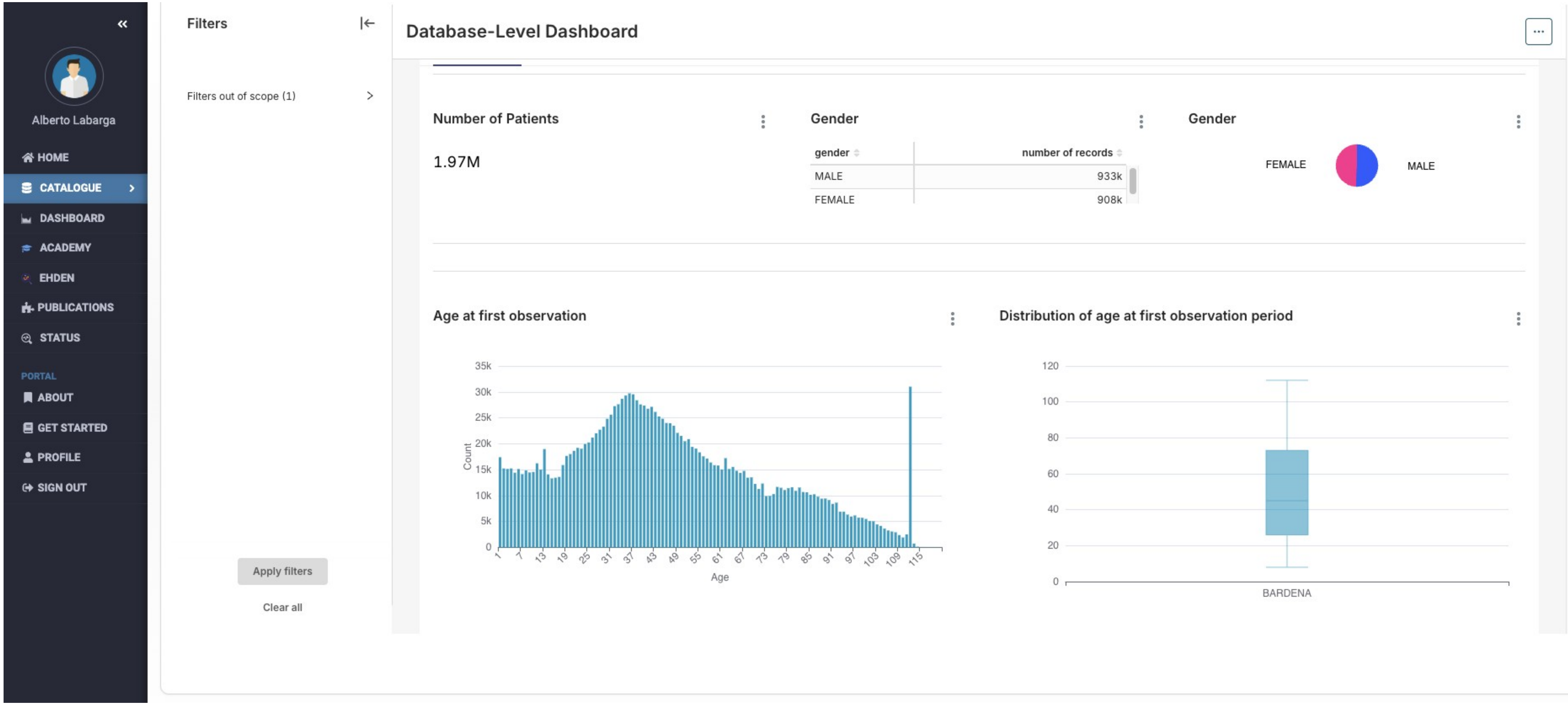
REUSABLE

Todos los datos están descritos con formatos estándard y tienen licencias de uso claramente descritas


Findable
Persistent Identifiers (PIDs)
iD
Rich metadata
m
Indexed data repositories
PIDs in metadata
iD
m
Accessible
Standard communications protocol
Open, free protocol
Authentication, where necessary
Metadata is always available
Interoperable
Vocabularies
Vocabularies are FAIR
Linked metadata
Reusable
Metadata have multiple attributes
Usage license
Provenance
Community standards



https://portal.ehden.eu/


Alberto Labarga
HOME
CATALOGUE
DASHBOARD
ACADEMY
EHDEN
PUBLICATIONS
STATUS
PORTAL
ABOUT
GET STARTED
PROFILE
SIGN OUT
Filters
Filters out of scope (1)
Apply filters
Clear all
Database-Level Dashboard
Number of Patients
1.97M
Gender
gender
number of records
MALE
933k
FEMALE
908k
Gender
FEMALE
MALE
Age at first observation
Count
Age
Distribution of age at first observation period
BARDENA

EHDS

Espacio Europeo de Datos de Salud

Data user

Any natural or legal person allowed to request access to electronic health data through EHDS for certain purposes

Data holder

Any natural or legal person holding non-anonymized electronic health data, which it has a right or obligation to process in its capacity as a controller, or anonymized electronic health data

Health Data Access Body

Public bodies established by MSs and EC to oversee and facilitate data access through EHDS

Data permit

Administrative decision issued by a health data access body to a data user to access and process specified electronic health data

Secure processing environment

Specialised infrastructure, offered by health data access body, to enable safe access and use of electronic health data electronic health data from **EHRs**;

healthcare-related **administrative data**, including dispensation, claims and **reimbursement** data

automatically generated personal electronic health data, through **medical devices**;

data from **wellness applications**;

other health data from medical devices.

population-based health data **registries** (public healt registries);

data from medical registries and **mortality registrie**

data from registries for medicinal products and medi devices;

health data from **biobanks** and associated databases.

human **genetic, epigenomic and genomic** data;

other **human molecular** data such as proteomic transcriptomic, metabolomic, lipidomic and other -omic data;

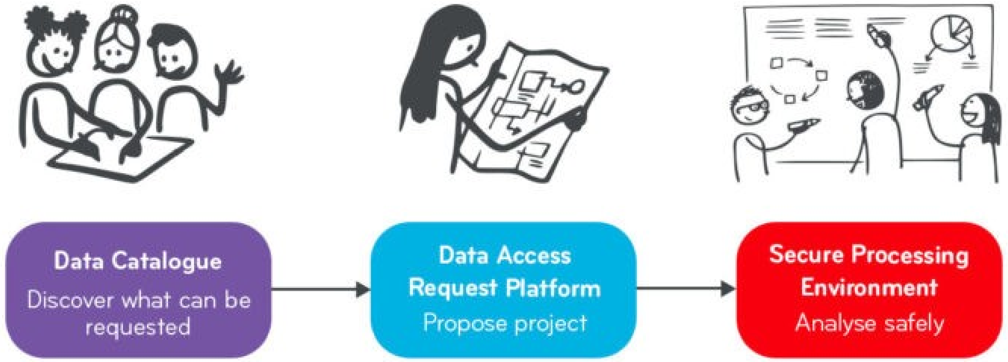
Data on factors impacting health, including **socio-economic, environmental and behavioural determinants** of health;

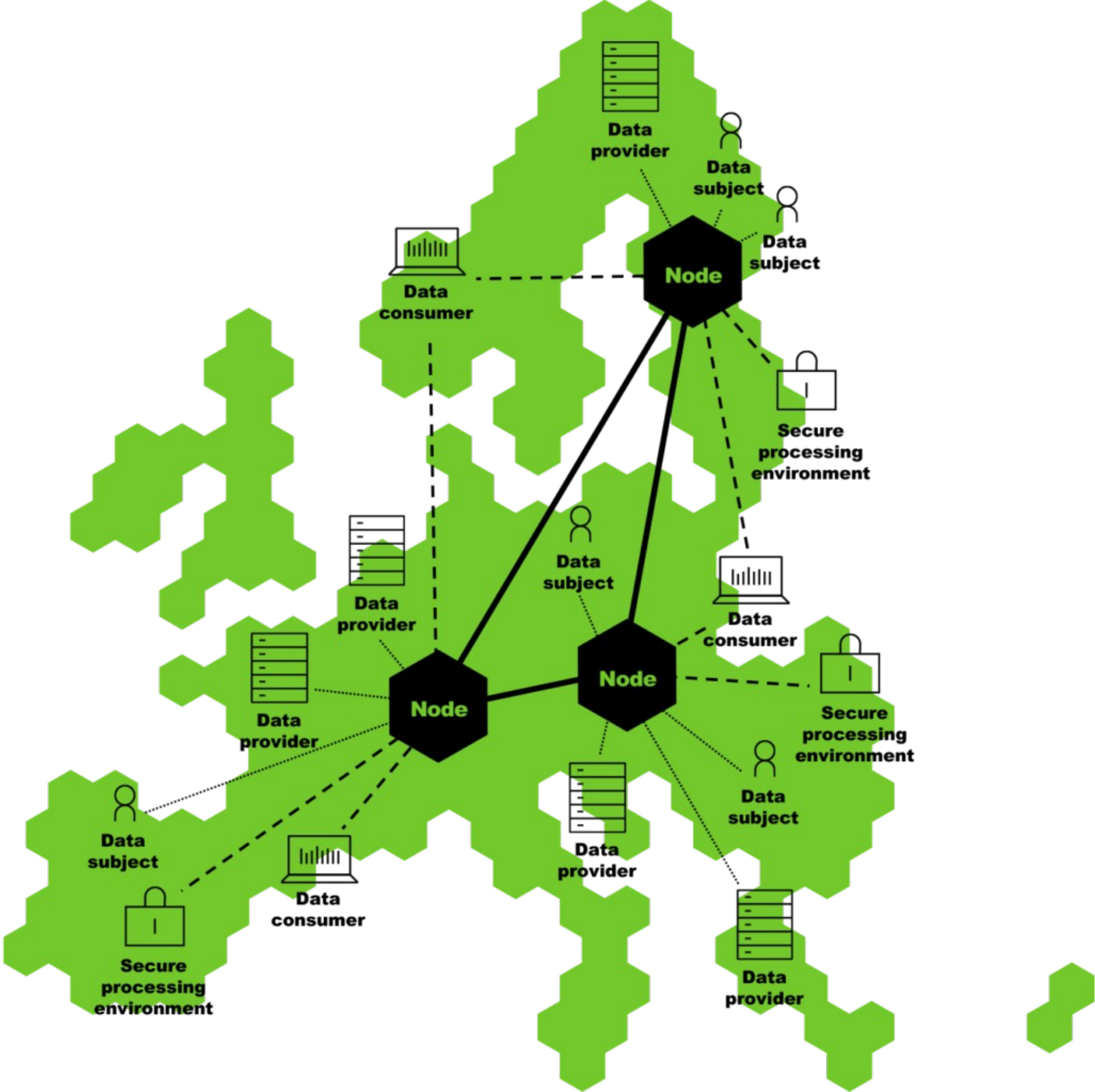
Aggregated data on **healthcare needs**, **resources** allocated to healthcare, the provision of and access to healthcare, healthcare expenditure and financing;

Pathogen data, impacting on human health

data from **clinical trials**, **clinical studies** and **clinical investigations** subject to Regulation (EU) 536/2014, Regulation [SOHO], Regulation (EU) 2017/745 and Regulation (EU) 2017/746, respectively;

data from **research cohorts, questionnaires** and surveys related to health, after the first publication of results


Data Catalogue
Discover what can be requested
Data Access Request Platform
Propose project
Secure Processing Environment
Analyse safely


Data provider
Data subject
Data subject
Data consumer
Node
Secure processing environment
Data provider
Data subject
Data consumer
Data provider
Node
Node
Secure processing environment
Data subject
Data provider
Data subject
Data consumer
Secure processing environment
Data provider

HEALTH DCAT-AP
CATÁLOGO DE DATOS

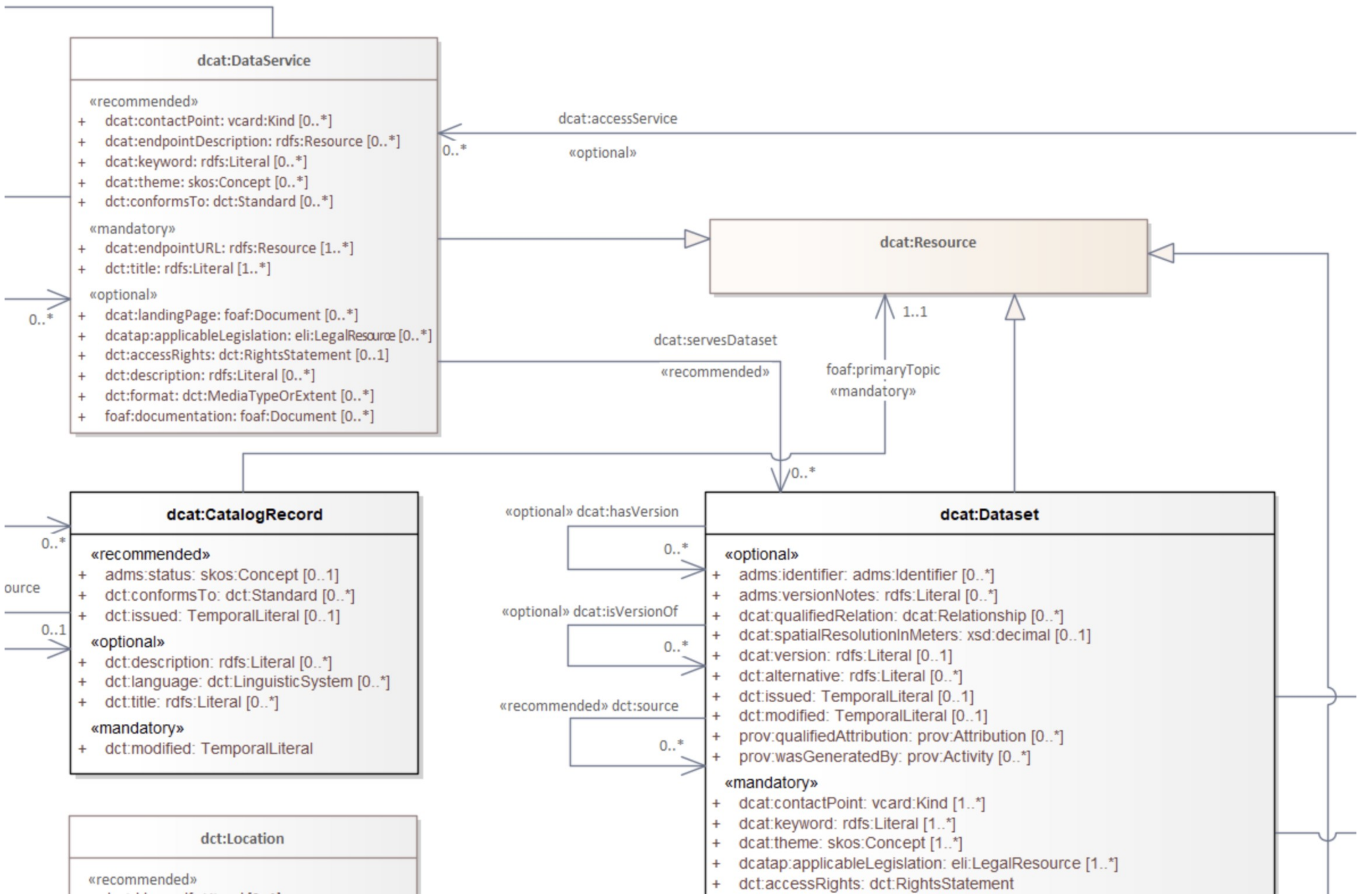
Catálogo (dcat:Catalog): El contenedor principal (ej. "Catálogo de Datos del Hospital La Paz").

Dataset (dcat:Dataset): La unidad conceptual de información (ej. "Registro de Pacientes con Diabetes Tipo 2 - 2020-2023"). No es el archivo Excel en sí, es el concepto.

Distribución (dcat:Distribution): La forma física o digital de acceder a los datos.

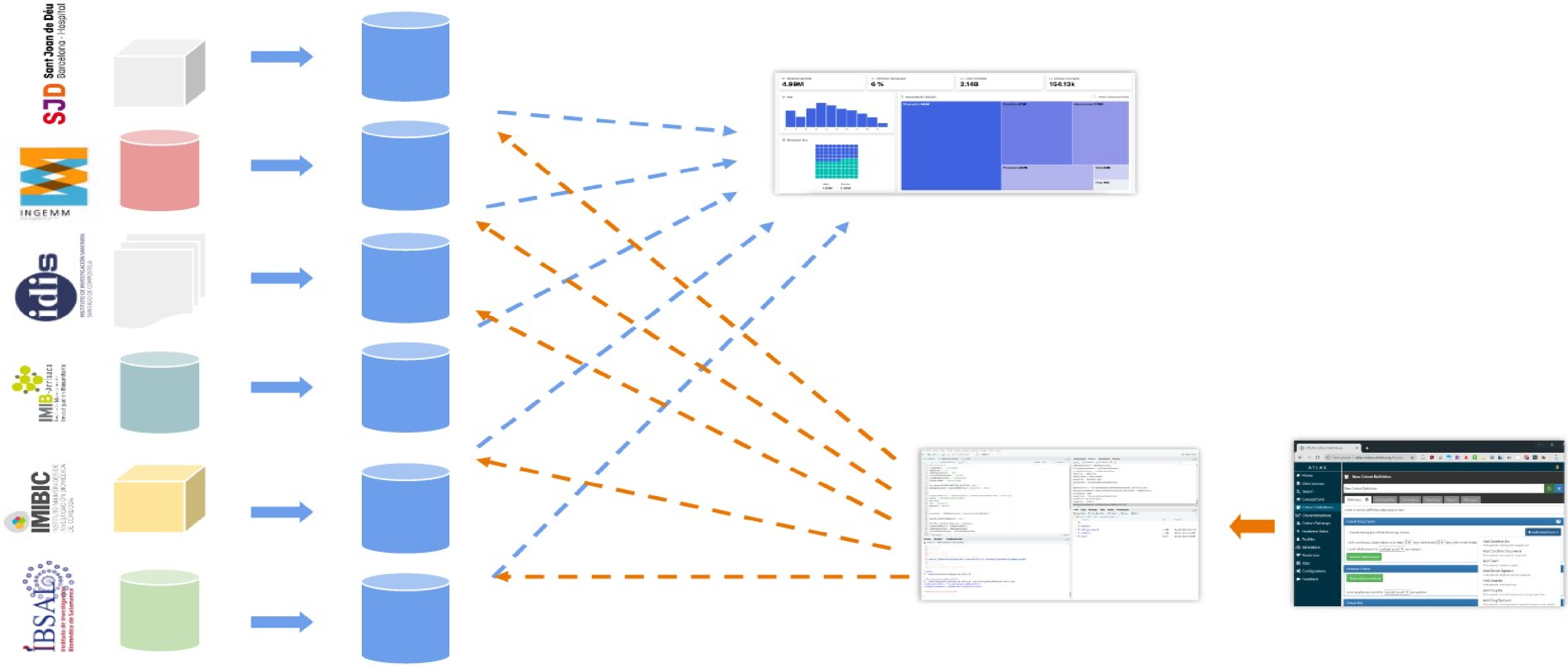
Servicio de Datos (dcat:DataService): Si los datos se sirven vía API (ej. FHIR API).

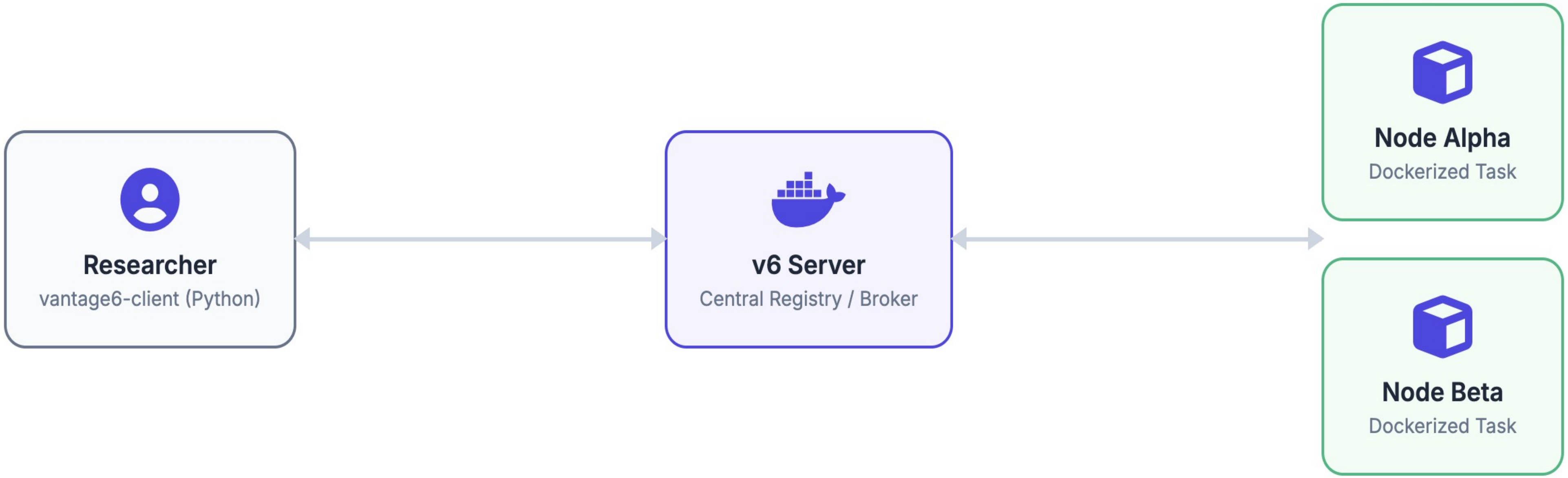
https://healthdataeu.pages.code.europa.eu/healthdcat-ap/releases/release-6/index.html



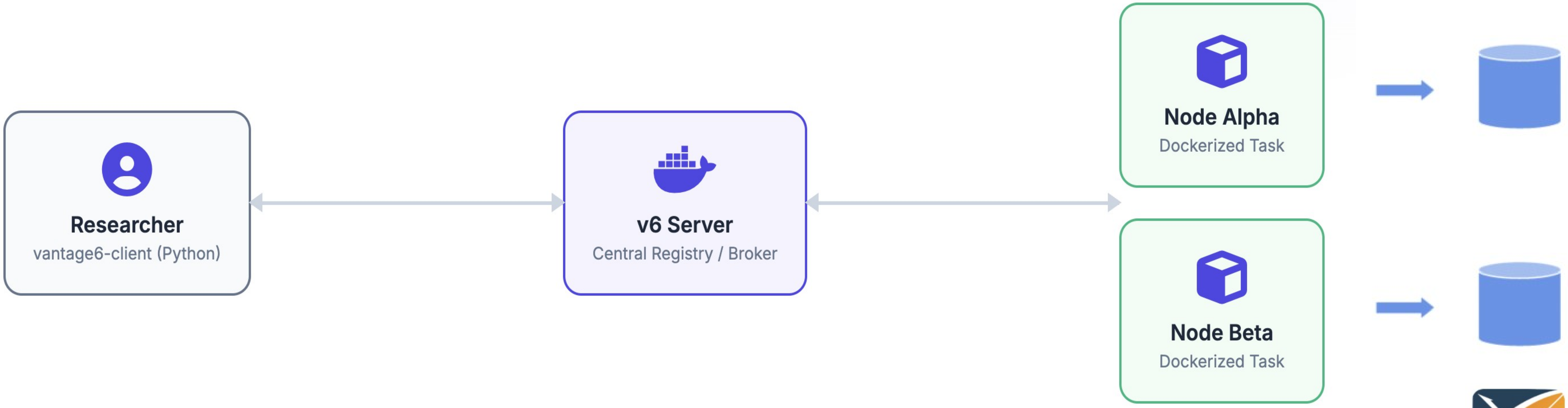
Federación

de datos


SJD Sant Joan de Déu
Barcelona · Hospital
INGEMM
idis
IMIB-Arrixaca
IMIBIC
IBSAL

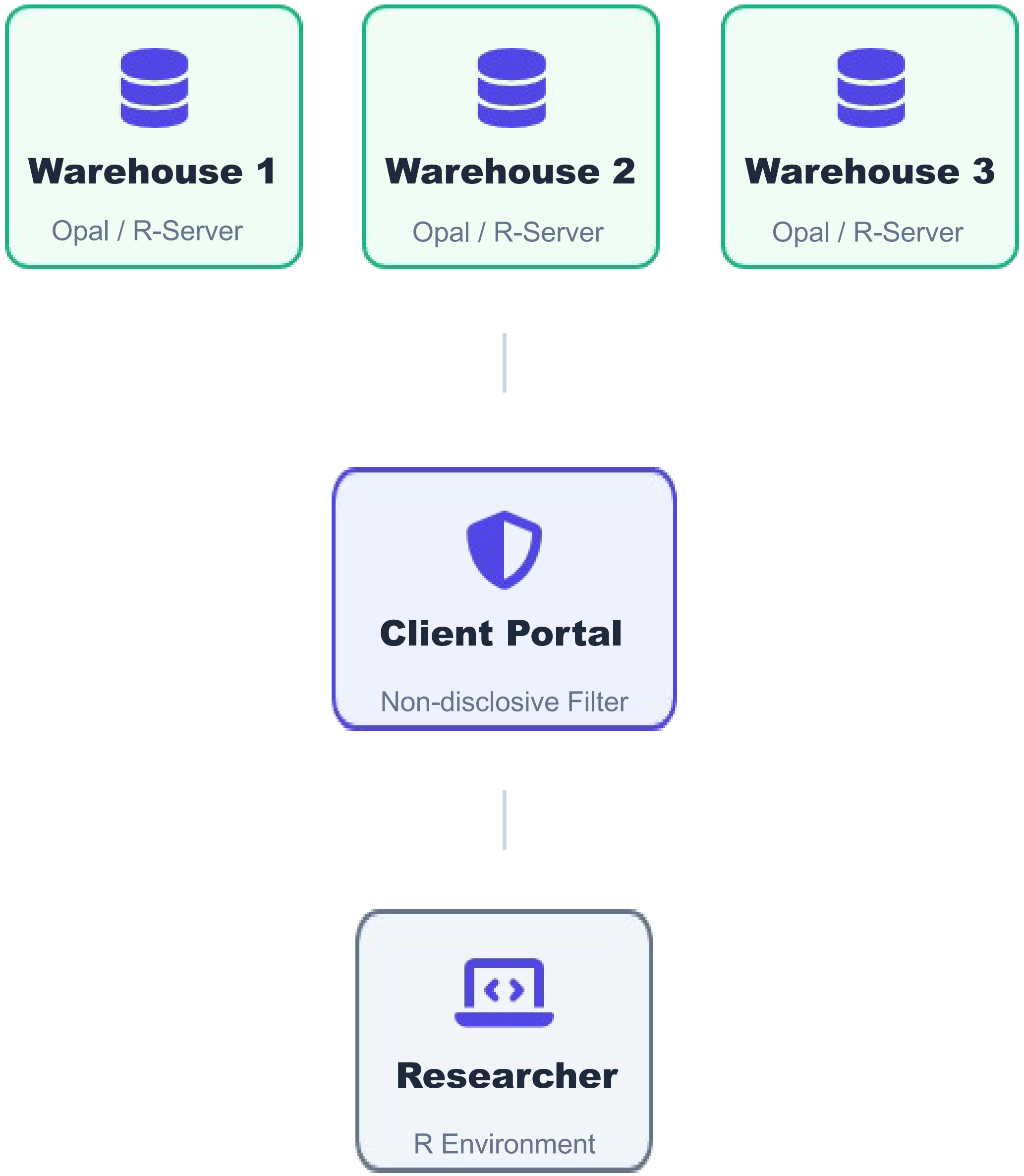

Researcher
vantage6-client (Python)
v6 Server
Central Registry / Broker
Node Alpha
Dockerized Task
Node Beta
Dockerized Task

VANTAGE



https://github.com/vantage6/python-ohdsi

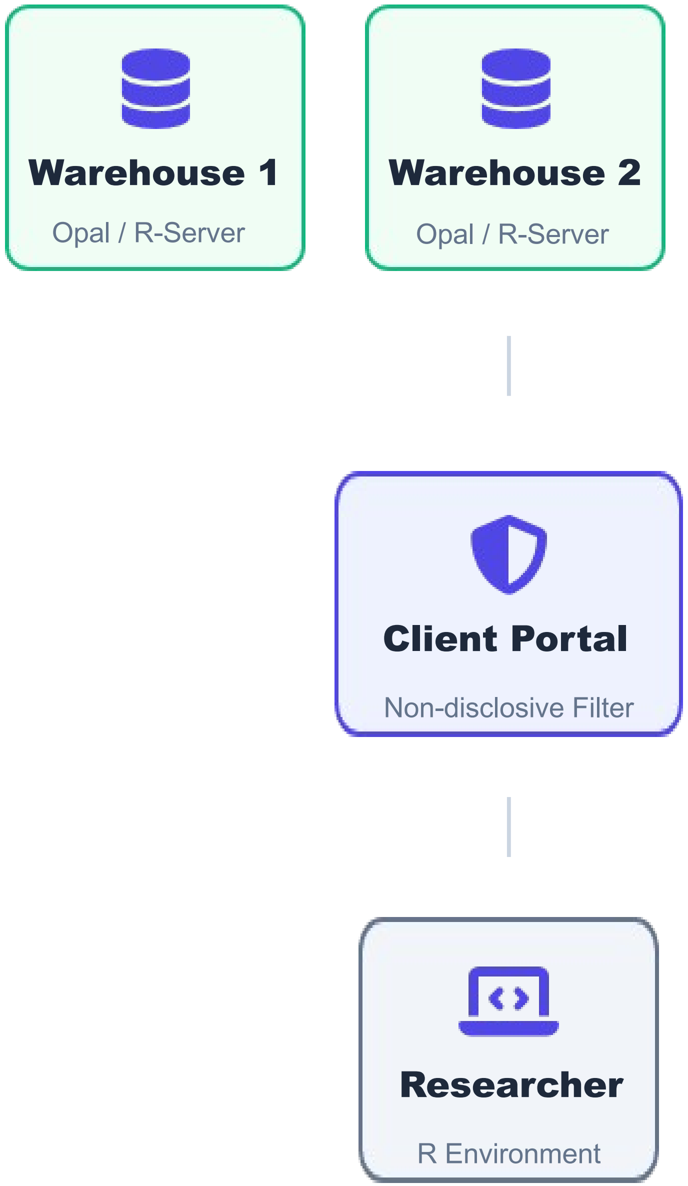
https://github.com/vantage6/ohdsi-api



DataSHIELD
Secure Bioscience Collaboration

https://datashield.org/

DataSHIELD
Secure Bioscience Collaboration



nfilter.tab

Blocks tables where any cell has fewer observations than a predefined threshold (e.g., < 3).

nfilter.subset

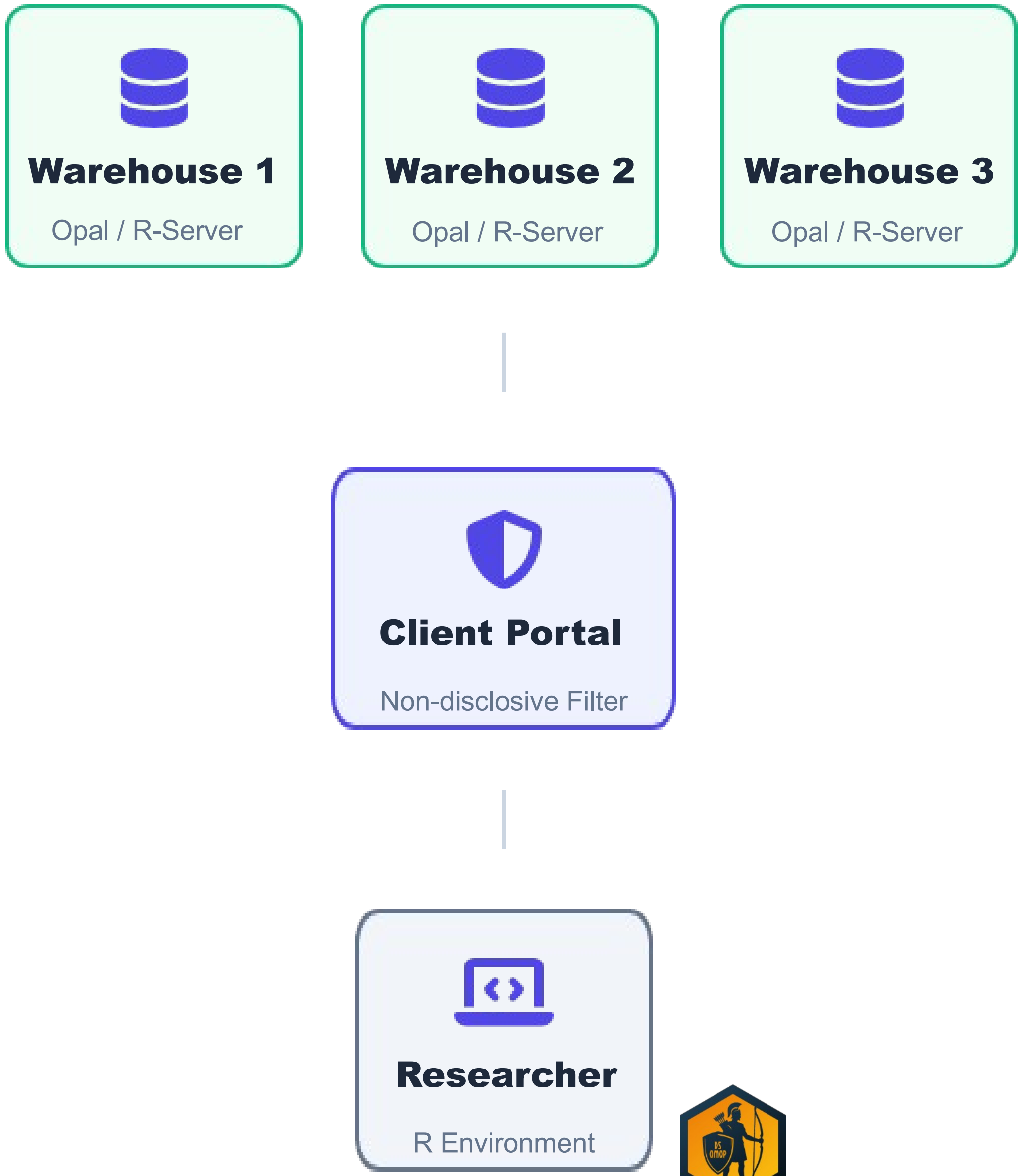
Ensures that any subset created from the data maintains a minimum required size (N).

nfilter.levels

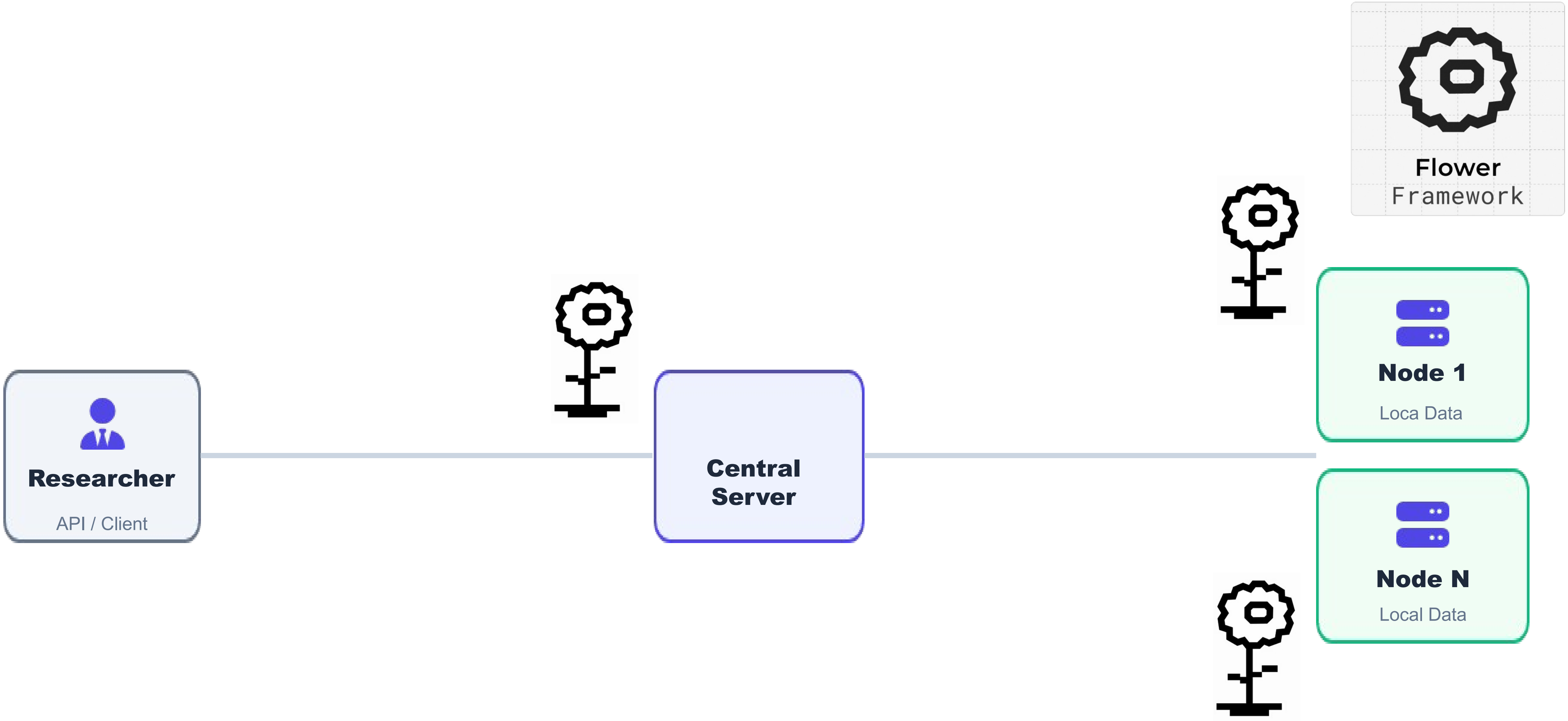
Restricts queries on variables with excessive unique levels that could identify individuals.

Visual Vetting

Plots are either aggregated or "jittered" to prevent visual identification of outliers.

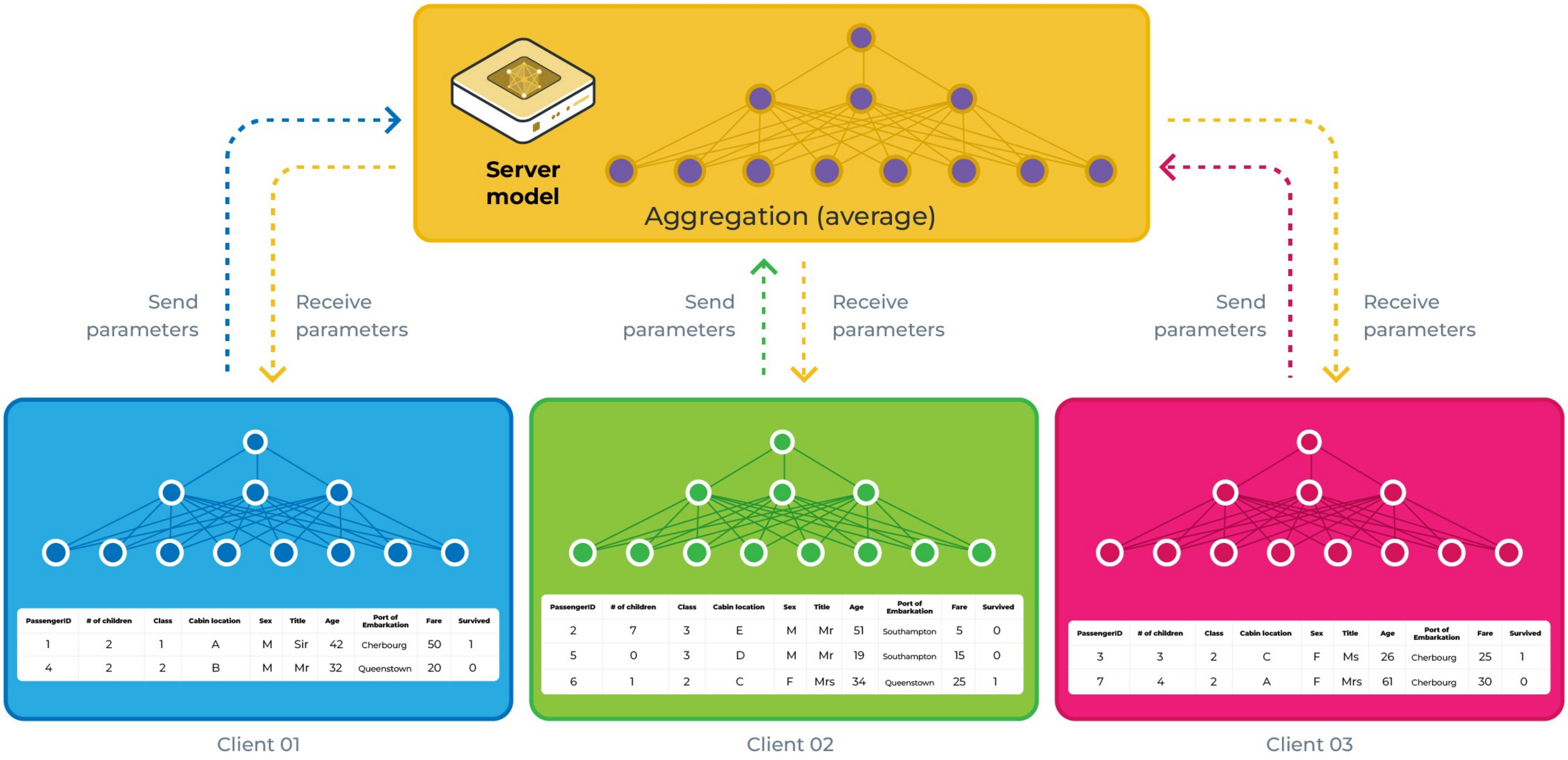


https://isglobal-brge.github.io/dsOMOP/



https://flower.ai/

Horizontal Federated Learning



Client 01

PassengerID	# of children	Class	Cabin location	Sex	Title	Age	Port of Embarkation	Fare	Survived
1	2	1	A	M	Sir	42	Cherbourg	50	1
4	2	2	B	M	Mr	32	Queenstown	20	0

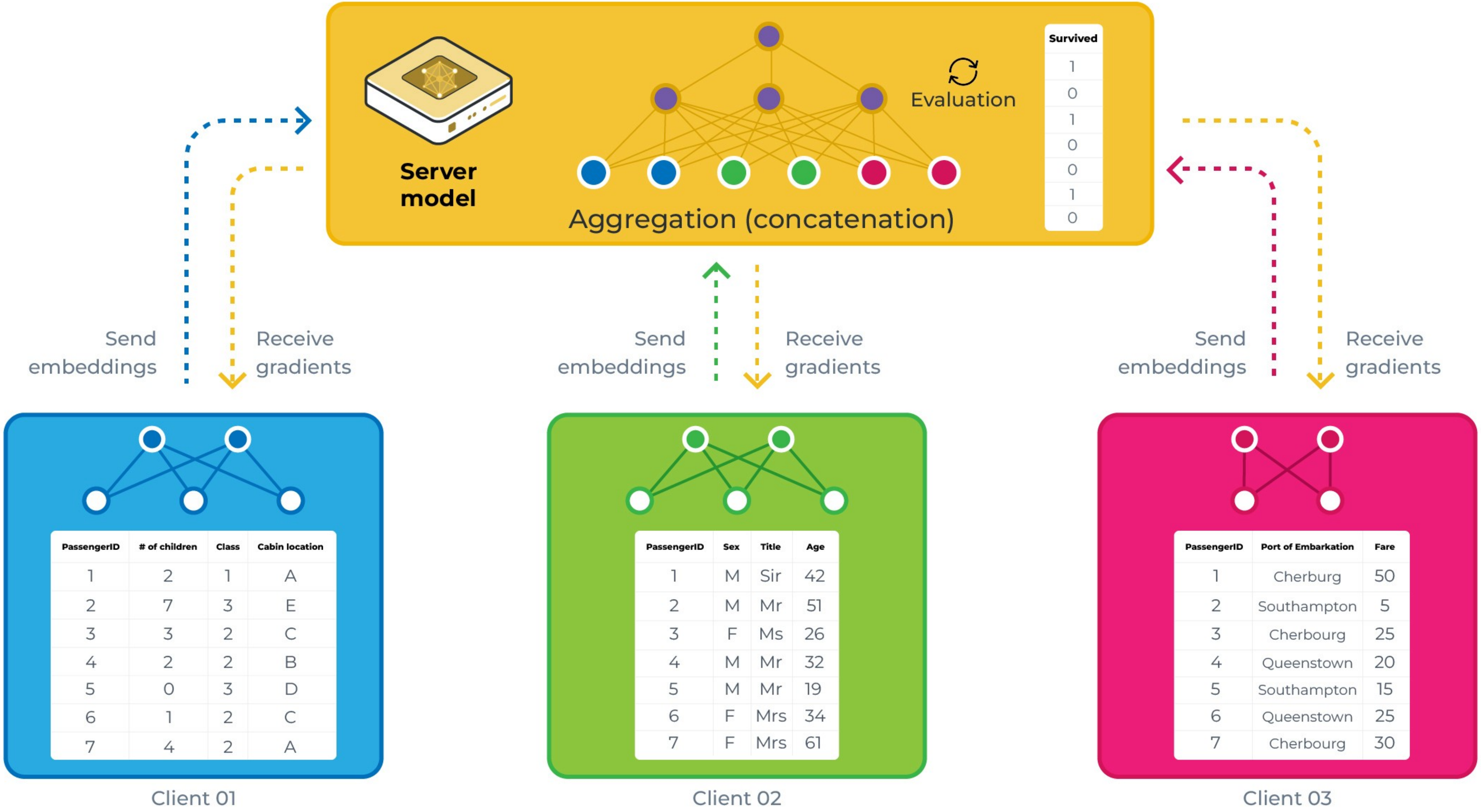
Client 02

PassengerID	# of children	Class	Cabin location	Sex	Title	Age	Port of Embarkation	Fare	Survived
2	7	3	E	M	Mr	51	Southampton	5	0
5	0	3	D	M	Mr	19	Southampton	15	0
6	1	2	C	F	Mrs	34	Queenstown	25	1

Client 03

PassengerID	# of children	Class	Cabin location	Sex	Title	Age	Port of Embarkation	Fare	Survived
3	3	2	C	F	Ms	26	Cherbourg	25	1
7	4	2	A	F	Mrs	61	Cherbourg	30	0

Vertical Federated Learning



Survived
1
0
1
0
0
1
0

PassengerID	# of children	Class	Cabin location
1	2	1	A
2	7	3	E
3	3	2	C
4	2	2	B
5	0	3	D
6	1	2	C
7	4	2	A

PassengerID	Sex	Title	Age
1	M	Sir	42
2	M	Mr	51
3	F	Ms	26
4	M	Mr	32
5	M	Mr	19
6	F	Mrs	34
7	F	Mrs	61

PassengerID	Port of Embarkation	Fare
1	Cherburg	50
2	Southampton	5
3	Cherbourg	25
4	Queenstown	20
5	Southampton	15
6	Queenstown	25
7	Cherbourg	30

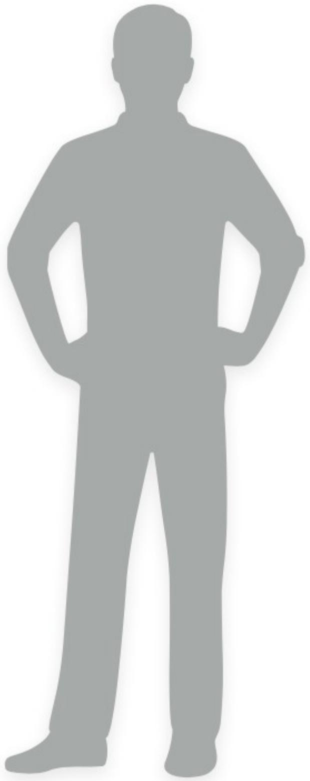
FEDERATED RAMEWORKS
UN RESUMEN

Feature	vantage6	DataSHIELD	Flower
Focus	Infrastructure / Docker	Statistics / Privacy	Deep Learning / Scale
Privacy	Isolated Containers	Real-time Filtering	Secure Aggregation
Language	Agnostic (Docker)	R Ecosystem	Python (generic)
Best For	Multi-centric ML	Epidemiology	Federated Learning

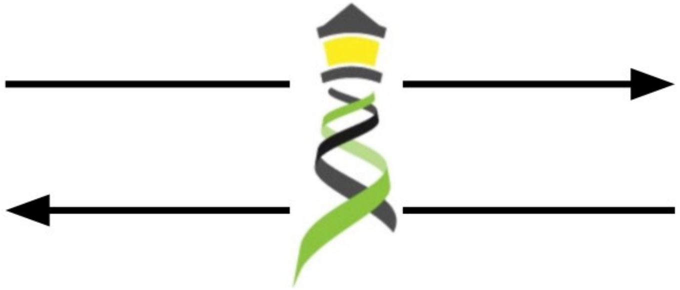
Federación

de datos genómicos

GA4GH Beacon

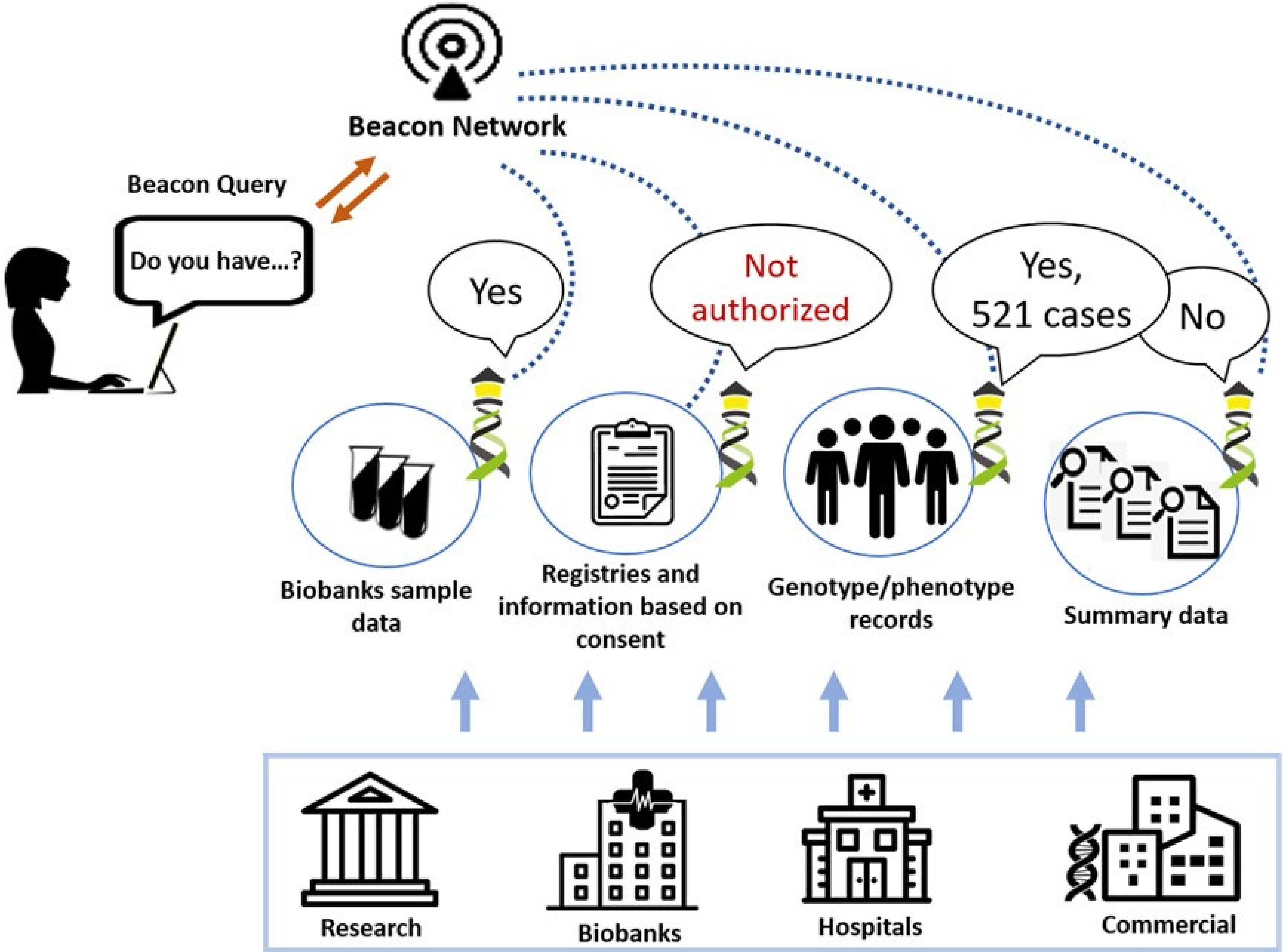


“Do you have a ‘C’ at chromosome 13 at position 32,936,732?”



“Yes” (or “no”)

https://github.com/ga4gh-beacon


Beacon Network
Beacon Query
Do you have...?
Yes
Not authorized
Yes, 521 cases
No
Biobanks sample data
Registries and information based on consent
Genotype/phenotype records
Summary data
Research
Biobanks
Hospitals
Commercial

Beacon REST API

While the full power of the Beacon API can be unlocked through the use of structured queries using JSON serialization ("POST" requests), the majority of common queries can be implemented through standard query URLs with parameters (GET queries).

Beacon API URL structure

Beacon REST paths in general follow the format

`__APIroot__/__entryType__/{id}/`

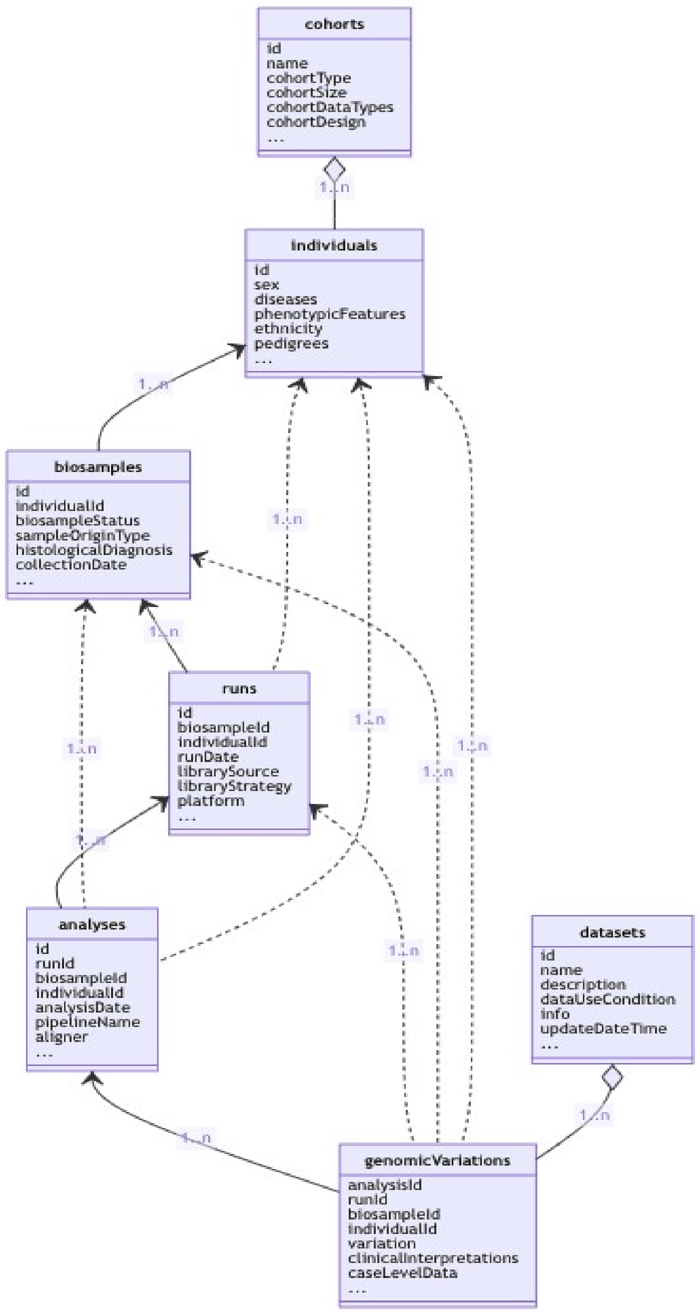
or

`__APIroot__/__entryType__/{id}/__requestedSchema__`

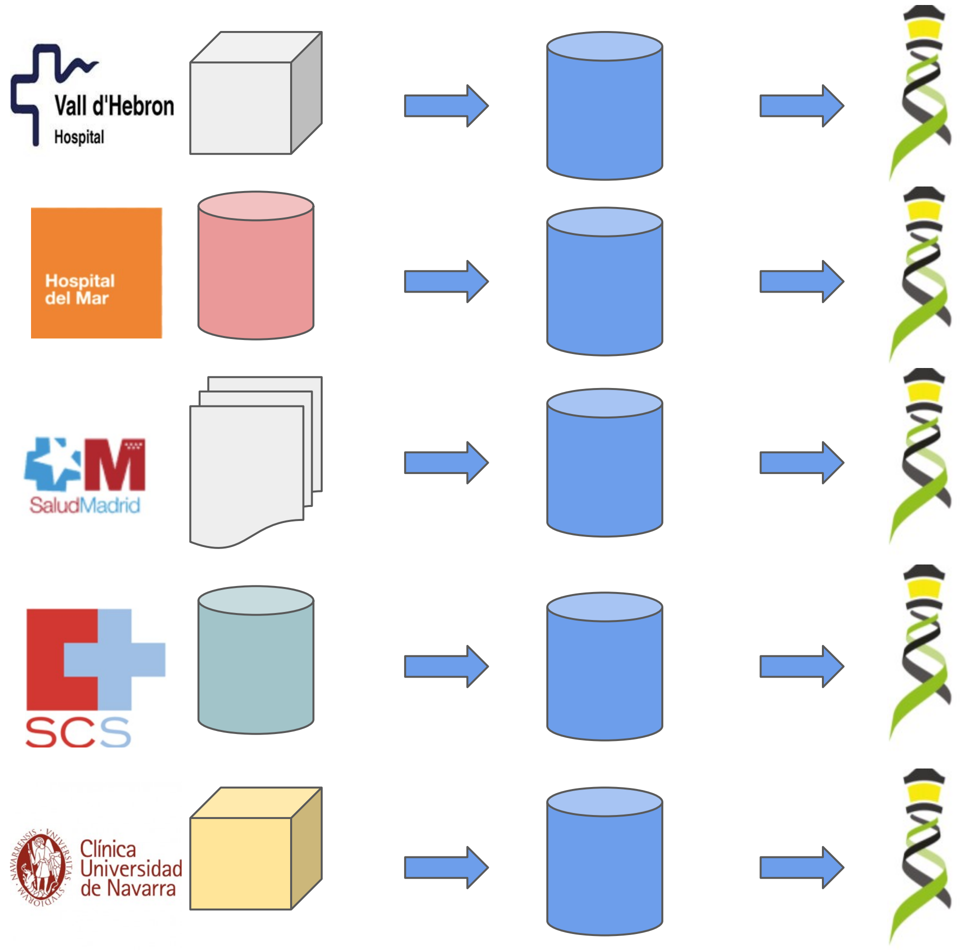
A typical example would e.g. the request to retrieve all genomic variants associated with a biosample

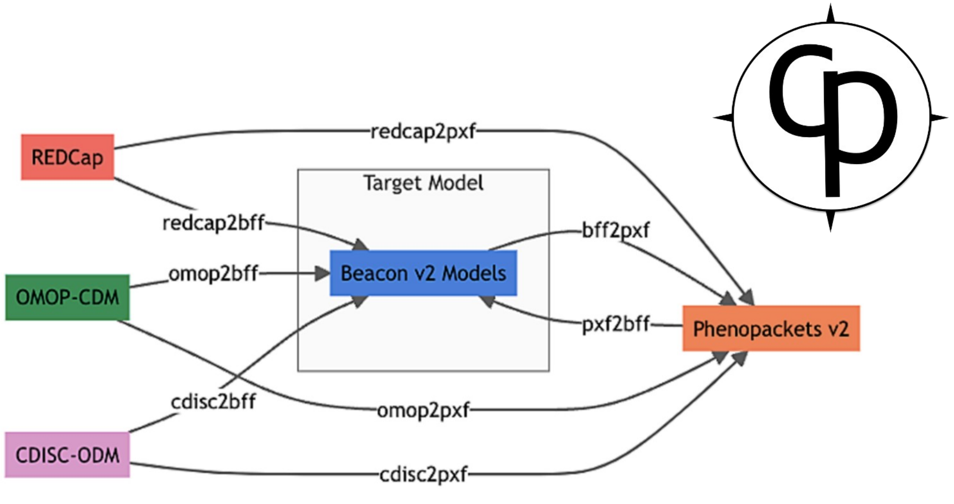
`https://example.com/beacon/api/biosamples/bios-st4582/g_variants`

REST Endpoint Definitions

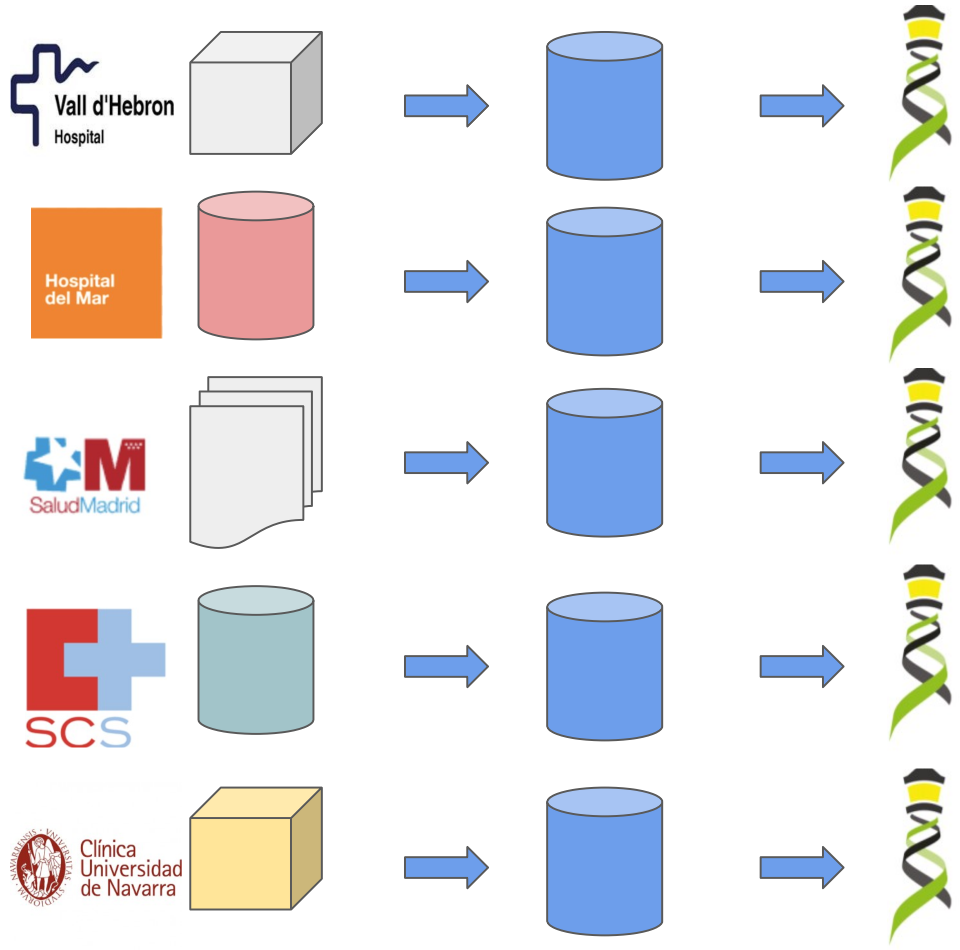


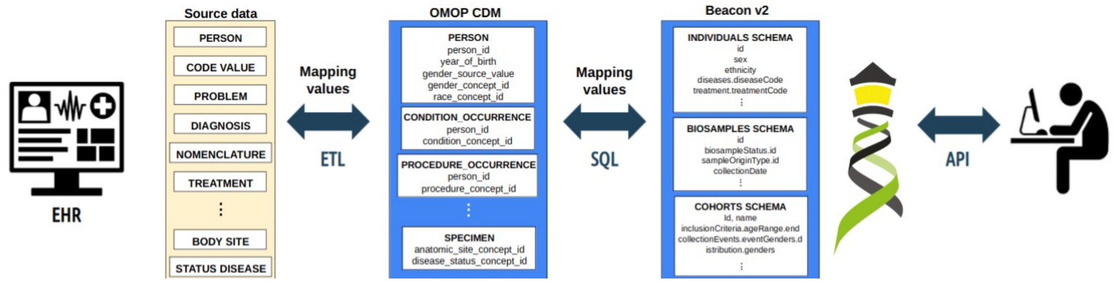

```json
{
    "$schema": "https://json-schema.org/draft/2020-12/schema",
    "id": "Ind001",
    "ethnicity": {
        "id": "NCIT:C43851",
        "label": "European"
    },
    "geographicOrigin": {
        "id": "GAZ:00002955",
        "label": "Slovenia"
    },
     "sex": {
        "id": "NCIT:C16576",
        "label": "female"
    }
    "diseases": [
        {
            "ageOfOnset": {
                "ageGroup": {
                    "id": "NCIT:C49685",
                    "label": "Adult 18-65 Years Old"
                }
            },
            "diseaseCode": {
                "id": "OMIM:164400",
                "label": "Spinocerebellar ataxia 1"
            }
        }
    ]
}
```

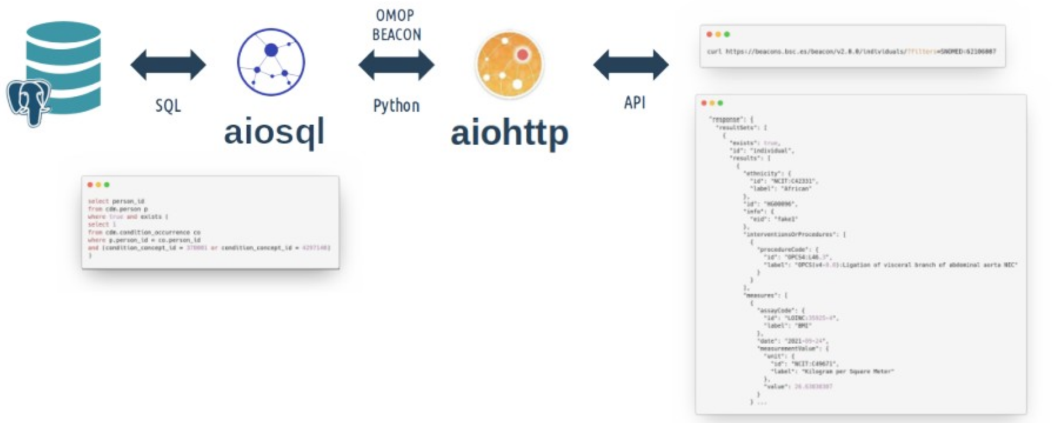




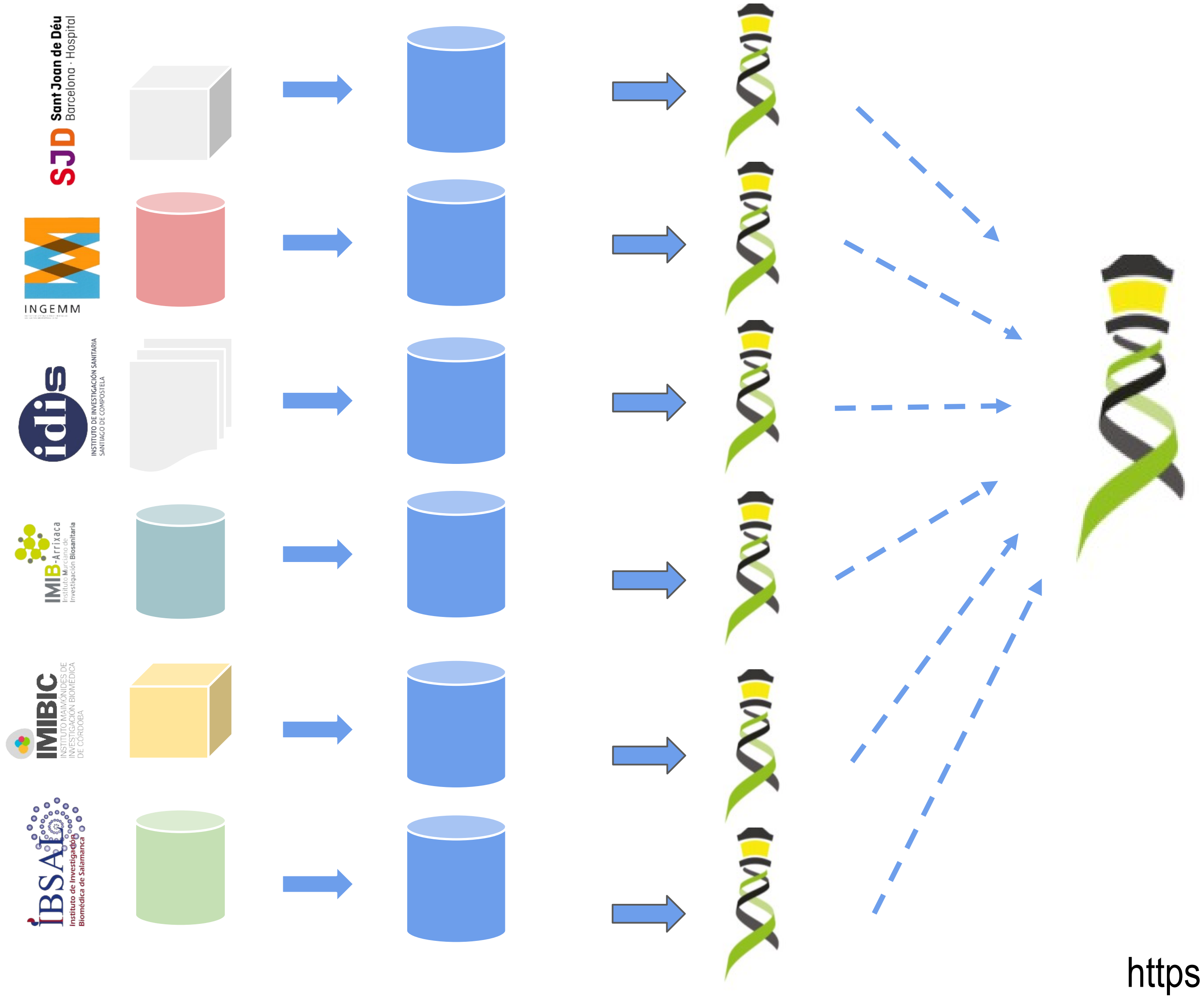
https://cnag-biomedical-informatics.github.io/convert-pheno/





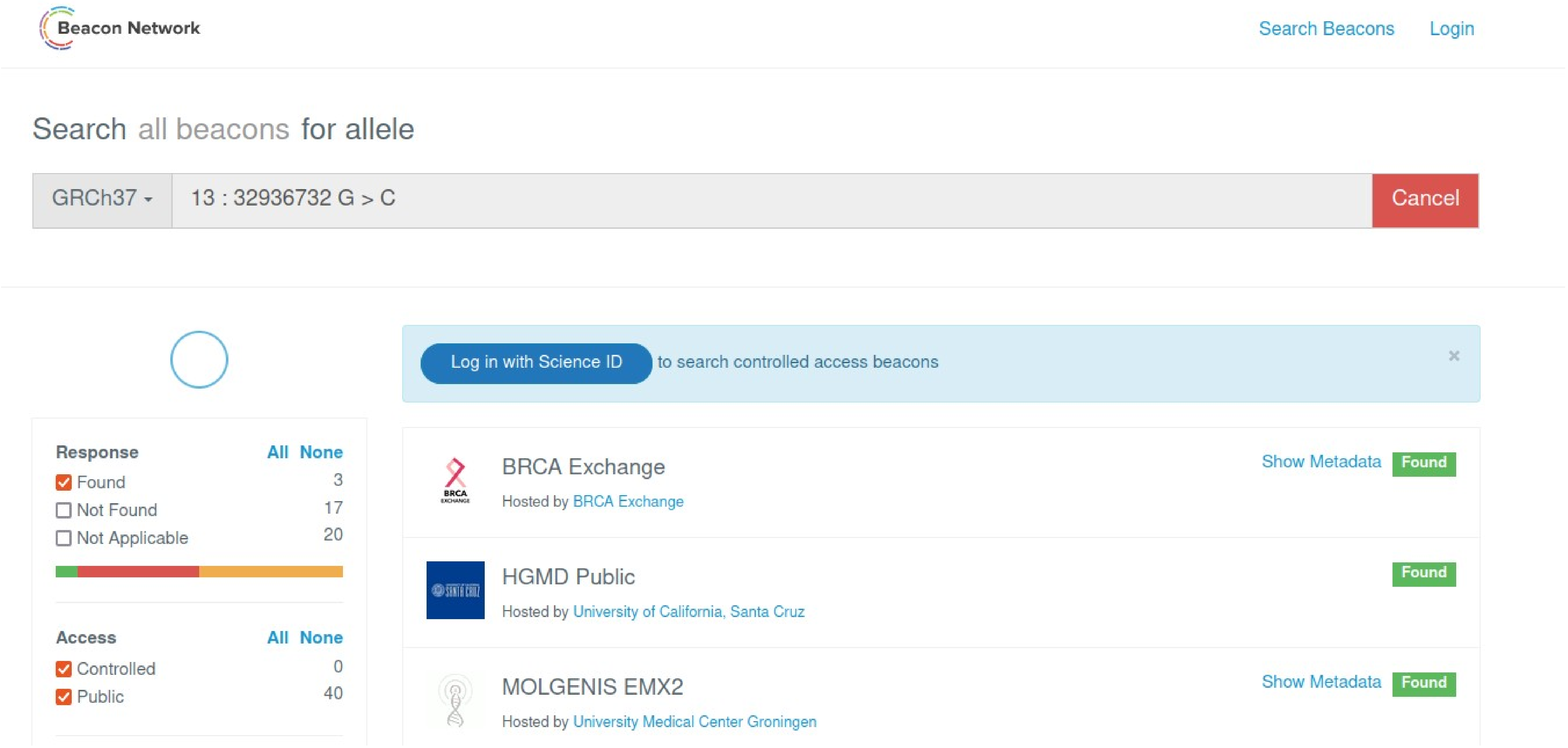


https://gitlab.bsc.es/impact-data/impd-beacon_omopcdm



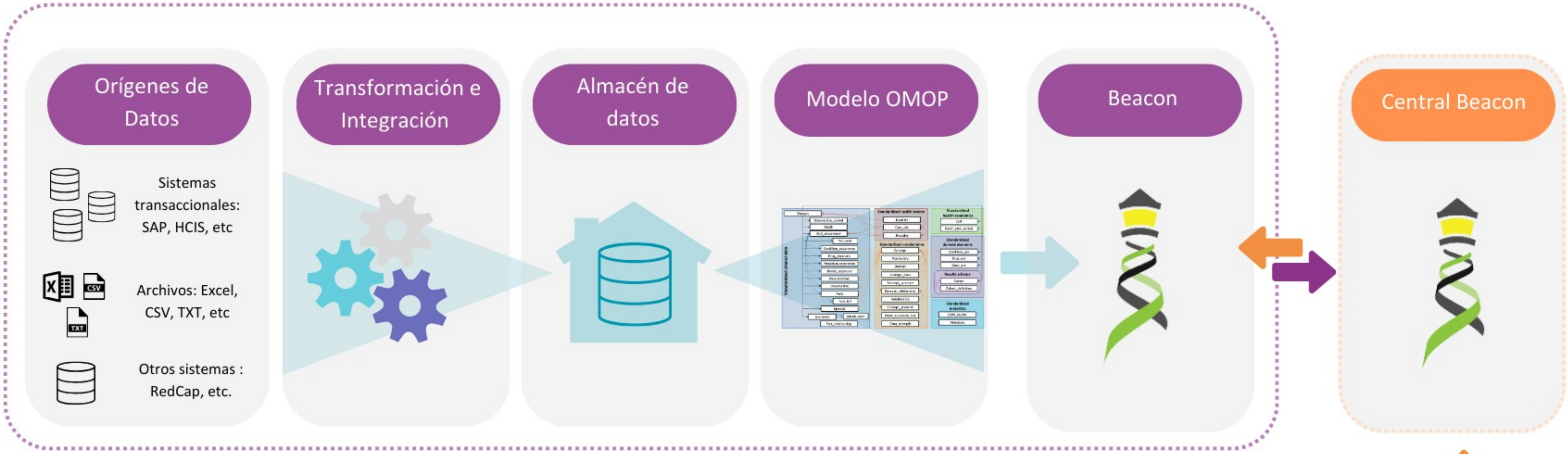
https://github.com/elixir-europe/beacon-network-docker

Beacon network



https://beacon-network-demo.ega-archive.org/

Modelo tecnológico de cada hospital

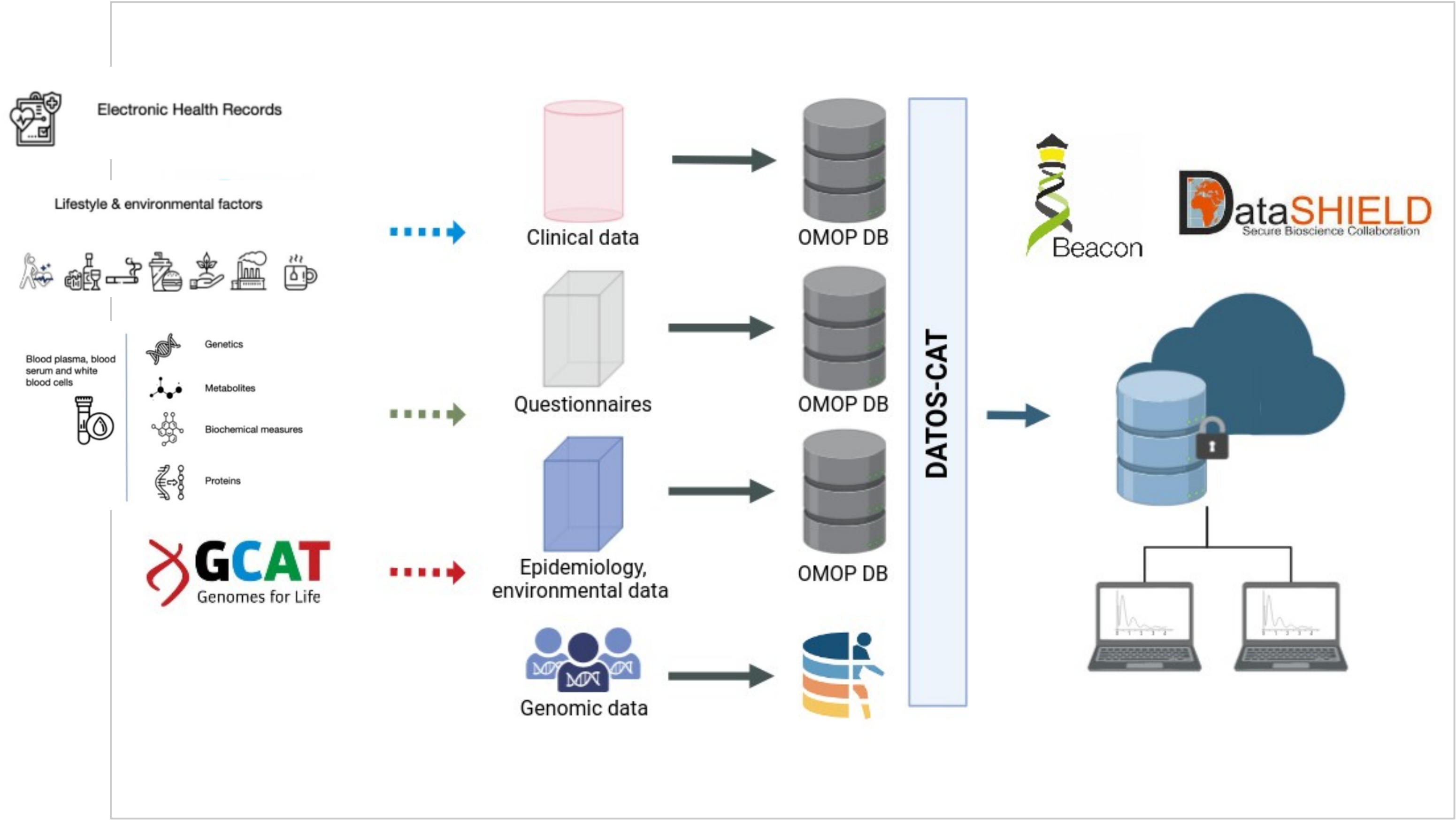


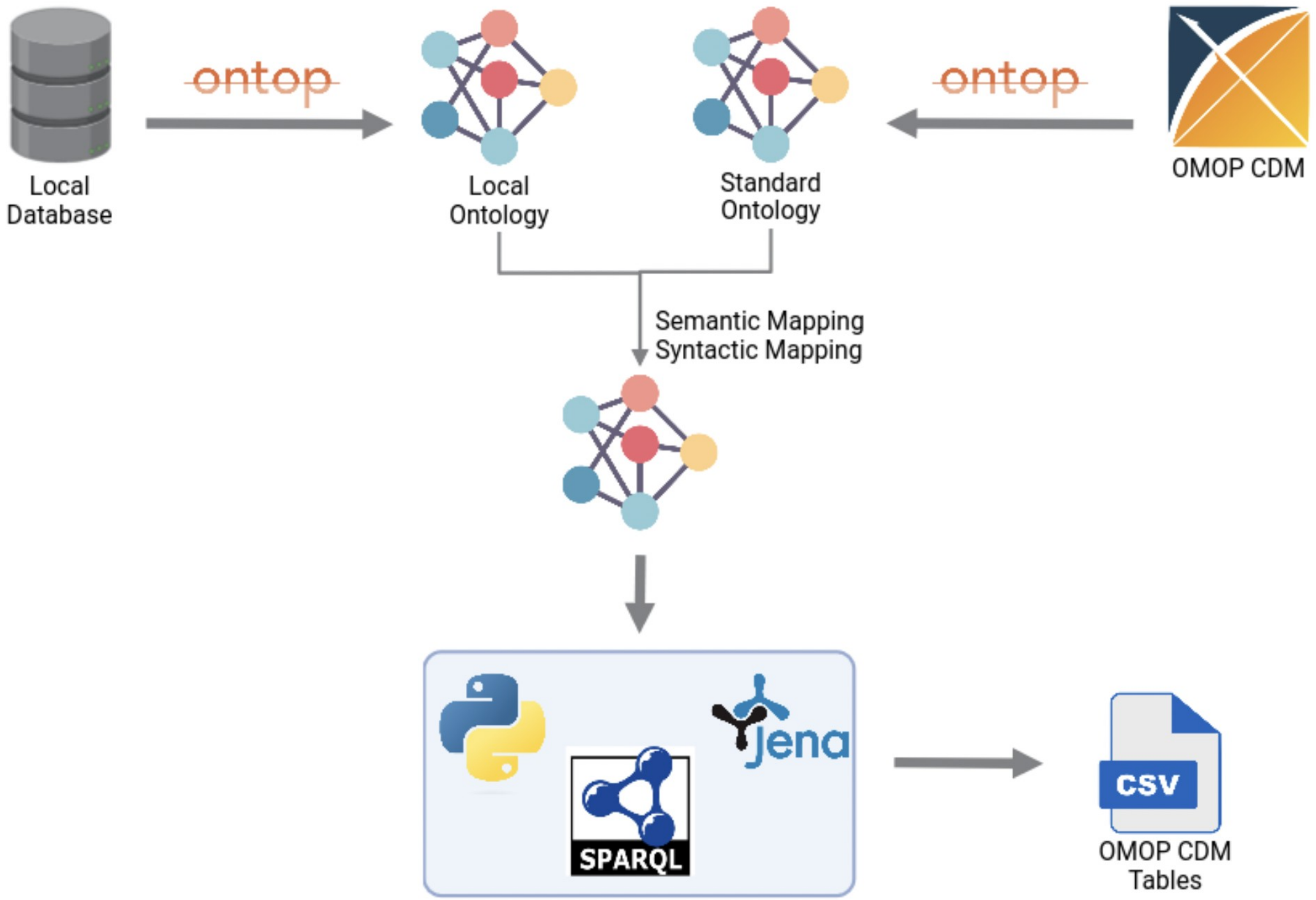
PROYECTO PILOTO ÚNICAS (IMPaCT3)

Implantación de una red pediátrica de medicina personalizada en enfermedades raras pediátricas

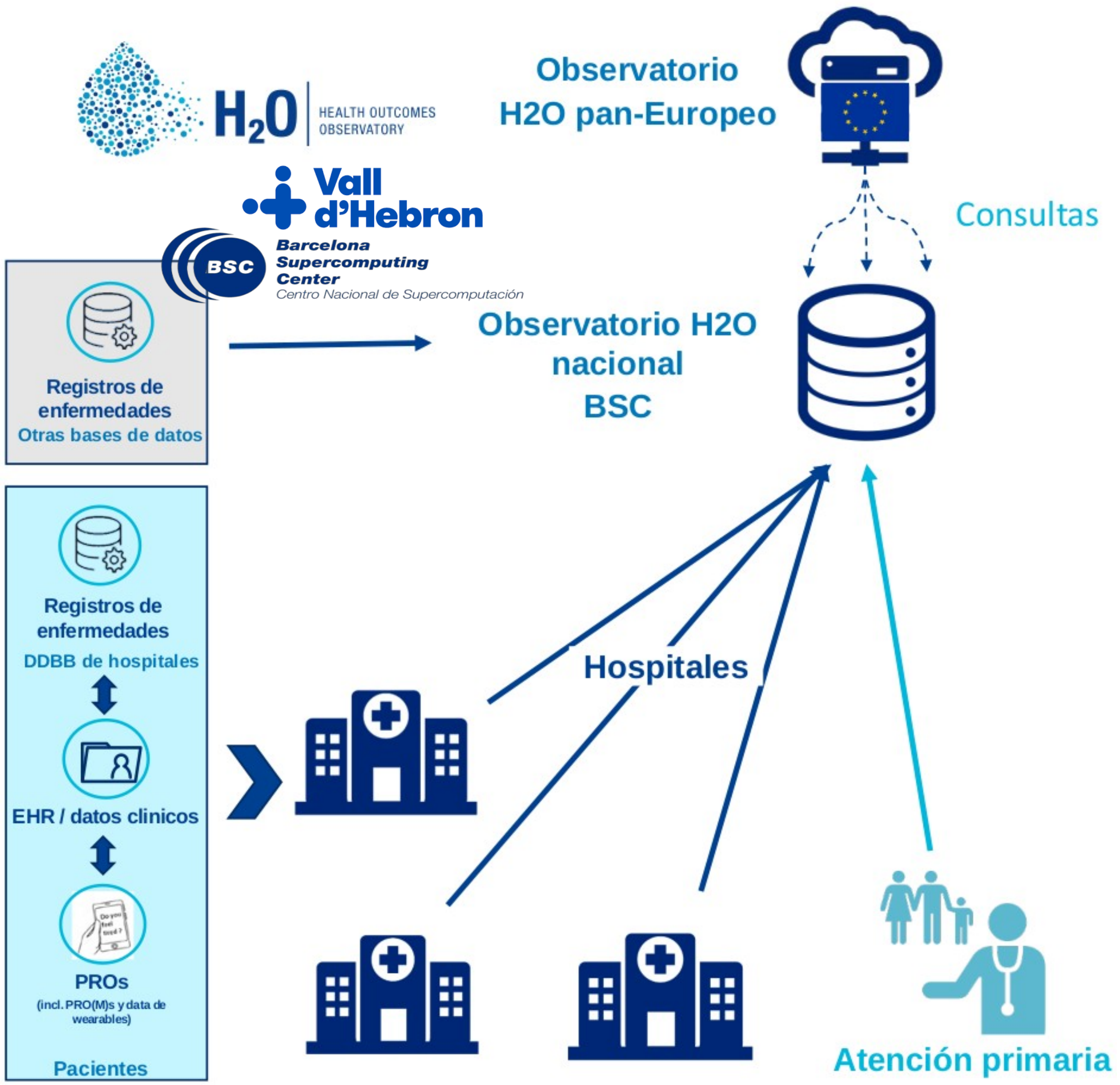
DATOS- CAT Implementación y análisis de bases de datos en medicina de precisión

El objetivo del proyecto es la **integración y publicación de datos** y metadatos desde distintos sistemas y repositorios a la base de datos poblacional catalana (**GCAT/COVICAT**). Los datos genómicos están disponibles en **EGA** y el resto de los datos se publicará de manera federada siguiendo los **principios FAIR** en formatos estándar como **OMOP-CDM, beacon**, etc





https://github.com/InformaticaClinica/OntoBridge



El **Observatorio de resultados de salud proyecto (H2O)** reúne a los sectores público y privado para crear un sistema de infraestructura y gobernanza de datos en toda Europa para **incorporar las experiencias y preferencias de los pacientes en las decisiones que afectan su atención** sanitaria individual y la de toda la comunidad de pacientes.

Los **Observatorios Nacionales** custodian los datos en nombre de los pacientes, quienes mantienen el control de sus propios datos que se utilizan para la atención individual y colectiva, la investigación o la evaluación de nuevas tecnologías y se conectan a un Observatorio paneuropeo para facilitar la **interoperabilidad** y **utilizar resultados centrados en el paciente a nivel regional, nacional, europeo y mundial**.

EL Observatorio Nacional en España está liderado por el **Hospital Vall d'Hebrón** con el soporte tecnológico del **BSC** y la colaboración del Instituto Catalá de la Salut

https://health-outcomes-observatory.eu/

Preguntas?

GRACIAS POR VUESTRA ATENCIÓN

OMOP-CDM

Publicación y explotación de datos